Convertible Bond Gamma Trading

Comprehensive Study Guide

For Quantitative Traders on the Convertible Bond Desk

Gamma Mechanics • Carry Decomposition • Implied Volatility Dynamics • Optimal Rebalancing • Non-Linear Costs • Liquidity Risk Premia • Leverage Effect Asymmetry

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Part I — Foundations of Convertible Bond Gamma Trading

1. The Convertible Bond as a Hybrid Instrument

A convertible bond is a corporate bond with an embedded call option allowing the holder to convert the bond into a fixed number of shares of the issuer's equity. This embedded optionality is the source of convexity, and convexity is the source of gamma. The gamma trade in convertible bonds is fundamentally a volatility arbitrage strategy that seeks to extract profit from the difference between the realized volatility of the underlying stock and the implied volatility priced into the convertible bond.

The trade is constructed by purchasing the convertible bond and simultaneously shorting a delta-equivalent number of shares of the underlying stock. Delta represents the sensitivity of the convertible bond's price to small changes in the stock price. When you hold a long convertible position and are delta-hedged, you are effectively isolated from directional stock movement (to a first-order approximation) and are instead exposed to the second-order effect: gamma.

2. Gamma Mechanics

Gamma measures how fast delta changes as the stock price moves. When you are long gamma, you benefit from stock price movement in either direction:

- **Stock rises:** Delta increases, meaning you were effectively underhedged — the convertible appreciated more than your short stock position lost.
- **Stock falls:** Delta decreases, meaning you were overhedged — the convertible lost less than your short position gained.

In both cases, you can rebalance your hedge (sell more stock if the stock rose, buy back stock if the stock fell) and lock in a profit. This is the act of "harvesting gamma."

KEY CONCEPT: Gamma P&L Formula

The profit from a single rebalancing event is approximately $0.5 \times \Gamma \times (\Delta S)^2$, where Γ is the dollar gamma and ΔS is the stock price change. This convexity-driven gain is always positive regardless of direction.

3. The Gamma-Theta Tradeoff

Gamma is not free. The cost of being long gamma is theta — the time decay of the embedded option. Each day that passes with the stock sitting still, the option component of the convertible loses a small amount of value. The fundamental equation of gamma trading is therefore a race between gamma gains (driven by realized stock volatility) and theta losses (driven by the passage of time).

- **Profitable:** Realized volatility exceeds the implied volatility embedded in the convertible's price.
- **Unprofitable:** Realized volatility disappoints, and theta bleeds dominate.

4. The Carry Component

Convertible bond gamma trades have a carry component that distinguishes them from vanilla options-based gamma strategies. Carry is composed of several elements:

Component	Description	Impact on P&L
Coupon Income	Interest payments from the bond	Positive
Short Rebate	Interest on proceeds from short stock sale	Positive
Funding Cost	Cost of financing the bond purchase via prime broker/repo	Negative
Stock Borrow Cost	Fee for borrowing shares to maintain short position	Negative

KEY CONCEPT: Positive Carry + Long Gamma

A positive-carry gamma trade means the trader is being paid to wait for volatility, rather than bleeding money while waiting. This is the convertible arbitrageur's ideal: earn income while you wait, and profit when the stock moves.

5. Total P&L Decomposition

The total P&L of a delta-hedged convertible bond position decomposes as follows:

- Gamma P&L: Rebalancing gains driven by realized volatility
- Carry: Net of coupon, short rebate, funding, and borrow costs
- Theta: Time decay of the embedded option (negative)
- Credit spread changes: Sensitivity to the issuer's credit quality
- Vega: Changes in implied volatility
- Higher-order effects: Rho (interest rates), correlation between credit and equity

Part II — Selecting Gamma-Rich Convertible Bonds

6. Moneyness and the Gamma Profile

Not all convertible bonds are created equal for gamma purposes. The convertible bond universe is segmented by moneyness — the relationship between the current stock price and the effective conversion price.

Moneyness	Delta Range	Gamma	Dominant Risk	Gamma Suitability
Deep OTM (“Busted”)	< 20%	Very Low	Credit	Unsuitable
Balanced / ATM	30–70%	Maximum	Equity Volatility	Ideal (“Sweet Spot”)
Deep ITM (“Equity-Like”)	> 80%	Low	Equity Directional	Unsuitable

KEY CONCEPT: The Sweet Spot
Delta in the 40–60% range is where gamma is at its maximum because delta is most sensitive to stock price changes. The embedded option is near the money, where the curvature of the option payoff is steepest. This is where the convertible bond offers the most convexity per unit of theta.

7. Ideal Candidate Characteristics

Beyond moneyness, several features make a convertible bond particularly attractive for gamma strategies:

Characteristic	Ideal Range	Rationale
Maturity	2–5 years	Balances sufficient gamma with manageable theta and credit exposure
Credit Quality	IG or strong BB	Minimizes probability that credit deterioration dominates P&L
Underlying Volatility	High and consistent realized vol	Strategy profits from movement; key metric is realized vs. implied spread
Liquidity	Tight bid-ask, sufficient size	Illiquid converts create hidden costs that erode gamma profits
Stock Borrow	Low borrow cost, easily borrowable	Borrow cost is a direct drag on carry; hard-to-borrow introduces squeeze risk
Call Protection	Long hard-call period	Call provisions truncate optionality and compress gamma
Structure	Simple conversion features	Complexity (CoCos, resets, make-whole) introduces model risk

Part III — Optimal Delta Hedge Rebalancing

8. The Theoretical Framework

In the idealized Black-Scholes world with continuous trading and no transaction costs, the delta hedge would be adjusted continuously and gamma P&L would exactly offset theta. In reality, trading is discrete and costly, so the trader must decide on a rebalancing rule.

The classical Whalley-Wilmott (1997) result shows the optimal no-trade band around the current delta is proportional to:

$$b^* \propto (3\lambda S / 2\Gamma)^{1/3}$$

where λ represents the proportional transaction cost (half the bid-ask spread), S is the stock price, and Γ is the gamma. The cube-root relationship means the optimal band is relatively insensitive to transaction costs (doubling the cost only widens the band by ~26%) and depends critically on gamma (high-gamma positions should be rebalanced more tightly).

9. Practical Rebalancing Approaches

Method	Description	When to Use
Delta Band	Rebalance when hedge ratio drifts by 3–5 delta points from theoretical	Default for most positions; calibrate to gamma and liquidity
Time-Based	Rebalance at fixed intervals (typically daily close)	Institutional baseline; supplement with intraday for large moves
Dollar-Gamma	Rebalance when dollar exposure from delta drift exceeds risk limit	Risk-budgeting approach tied to position size
Vol-Regime Adaptive	Increase frequency in high-vol regimes, decrease in low-vol	Sophisticated; captures more gamma when available, saves costs when not

10. Rebalancing P&L Tradeoff

Each rebalance captures a gamma profit of approximately $0.5 \times \Gamma \times (\Delta S)^2$ but incurs a transaction cost of approximately $\lambda \times |\Delta\text{shares}| \times S$.

- **Too frequent:** Captures all gamma but pays enormous cumulative costs (profit scales with $(\Delta S)^2$ which is tiny for small moves, cost scales linearly with number of trades).
- **Too infrequent:** Captures only a fraction of gamma and bears significant directional risk between rebalances.

KEY CONCEPT: Empirical Guideline

For typical convertible bond gamma trades in liquid stocks, rebalance once daily as a baseline, with supplementary intraday rebalances when the stock moves by more than 1.5 to 3 standard deviations of its daily return.

11. Asymmetric Rebalancing Considerations

A sophisticated refinement involves treating upside and downside moves differently. For a typical balanced convert, gamma tends to be slightly higher on the downside (bond floor creates steeper curvature) and may decrease on the upside (approaching call price). This suggests tighter rebalancing bands on the downside and wider bands on the upside in some cases.

Furthermore, a sharp stock decline may coincide with credit spread widening, causing the bond floor to drop and reducing effective gamma. This negative convexity in crisis scenarios means the gamma trade may not perform as well in large down-moves as a simple model suggests.

Part IV — Key Metrics and Portfolio Management

12. Critical Metrics to Monitor

12.1 Implied-Realized Volatility Spread

The single most important metric. Compare the convertible's implied volatility (backed out from market price using a pricing model) against trailing realized volatility over 10, 20, 30, and 60-day windows, plus forward-looking estimates from listed options or GARCH models.

- **Favorable:** Realized > implied by an increasing margin.
- **Warning:** Implied catching up to or exceeding realized signals shrinking edge.

12.2 Gamma-to-Theta Ratio and Breakeven Move

The daily breakeven move is approximately:

$$\text{Breakeven} = \sqrt{2\theta / \Gamma}$$

where θ is daily theta and Γ is dollar gamma. If the stock's expected daily move (based on realized volatility) comfortably exceeds this breakeven, the trade is attractively positioned.

12.3 Delta Drift and Hedge Ratio

Track in real time the current delta vs. the actual hedge in place. When deviation exceeds the predefined band, trigger a rebalance. Also track how rapidly delta is changing to anticipate future rebalancing frequency.

12.4 Credit Spread and Credit Risk

Monitor CDS spreads, asset swap spreads, fundamental metrics (leverage, earnings, liquidity), and rating agency actions. Credit deterioration reduces the bond floor, pushes the convert toward busted territory, and erodes the gamma profile.

12.5 Stock Borrow Cost and Availability

A sudden spike in borrow costs can turn a positive-carry trade negative. Maintain prime broker relationships and monitor borrow markets continuously.

12.6 Vega and Implied Volatility

Mark-to-market P&L is affected by implied volatility changes through vega, even if realized volatility remains favorable.

12.7 Parity, Premium, and Bond Floor Distance

For gamma trading, you want the convert in the balanced zone where parity is close to the bond price, conversion premium is moderate (10–30%), and there is meaningful distance to the bond floor.

13. Decision Framework

Signal	Action
Implied-realized spread favorable, trade	Maintain position, continue systematic rebalancing

Signal	Action
performing	
Spread narrowing	Reduce size or tighten rebalancing bands to preserve gains
Credit deterioration	Reduce or exit — credit risk is left-tail and can overwhelm gamma profits
Convert drifts to busted territory	Exit unless specific credit recovery thesis exists
Convert deeply ITM and callable	Exit or reduce — gamma compressed by call provision
Borrow costs spike	Recalculate carry; exit if carry too negative to justify gamma exposure

Part V — Cheapness Assessment

14. Theoretical Value Analysis

Price the convertible using a quantitative model (binomial/trinomial tree, finite-difference PDE, or Monte Carlo) incorporating the stock price process, credit risk (hazard rate or credit spread), interest rates, dividends, and structural terms. If market price < theoretical value, the convert is cheap (market is underpricing the embedded option).

The key input is the volatility assumption. Common approaches include trailing realized vol, listed options implied vol at comparable moneyness/maturity, or a blended estimate. If cheap even at conservative (lower) vol, the trade is particularly attractive.

15. Implied Volatility Extraction and Comparison

Extract the convertible's implied volatility by solving for the vol that makes model price equal market price. Compare against:

- **Listed options IV:** If convert implies 25% but listed options at similar maturity/moneyness imply 35%, the convert is cheap by 10 vol points. Adjust for maturity, moneyness, dividends, credit.
- **Historical realized vol:** If stock has realized 40% and convert implies 25%, there is a large cushion assuming realized vol doesn't collapse.
- **Forward-looking estimates:** GARCH models, options term structure, fundamental catalysts.

16. Practical Cheapness Indicators

- New issuance at conversion premium and implied vol below comparable recent deals (supply pressure).
- Orphaned converts dropped from indices or abandoned by large holders (technical selling).
- Broad market outflows from convertible funds (de-leveraging creates favorable entry points).
- Implied volatility term structure dislocations around catalysts.

17. Pitfalls in Cheapness Assessment

Apparent cheapness may be illusory when:

- The model is not correctly capturing credit risk — the bond floor is lower than assumed.
- Structural features (contingent conversion, mandatory conversion, make-whole) reduce actual optionality vs. vanilla.
- High dividend yields cause models to underestimate dividend impact on option value.

KEY CONCEPT: Robustness Principle

The most robust cheapness assessments combine multiple approaches — model-implied cheapness, implied-vol-versus-listed-options, implied-vol-versus-realized, and understanding of technical/fundamental drivers — with intellectual humility about model limitations.

Part VI — Implied Volatility Dynamics and the Gamma Trade

33. Vega Exposure in the Delta-Hedged Convertible

A delta-hedged convertible bond position is not a pure realized-volatility bet. It also carries significant vega exposure — sensitivity to changes in the implied volatility of the embedded option. While gamma P&L accrues from actual stock movements (realized volatility), the mark-to-market value of the convertible changes with the market's expectation of future volatility (implied volatility). This distinction is critical: a gamma trader can be harvesting gamma profitably from realized vol while simultaneously losing money on mark-to-market because implied vol is declining.

The vega of a convertible bond is largest when the convert is at the money (delta 40–60%) and declines as the convert moves toward busted or equity-like territory. This means the positions most attractive for gamma are also the ones most exposed to implied volatility swings. Understanding and managing this dual exposure is what separates sophisticated practitioners from mechanical delta-hedgers.

34. When Implied Volatility Rises

34.1 Direct Price Effect

An increase in the implied volatility of the convertible's embedded option raises the convertible's theoretical value. The option component becomes more valuable, and the convert's price rises. For a delta-hedged position (long convert, short stock), this is an unambiguous mark-to-market gain on the convertible leg, with no offsetting loss on the short stock (which has no vega). The P&L contribution is approximately:

$$\Delta P\&L_vega \approx Vega \times \Delta\sigma_implied$$

This can be substantial. For a balanced convertible with a vega of $2,000 per vol point, a 3-point increase in implied vol generates $6,000 of mark-to-market profit independent of any gamma harvesting.

34.2 Effect on Gamma and Theta

Rising implied volatility also affects the convertible's gamma and theta, though the direction depends on moneyness:

- **At-the-money converts:** Higher implied vol slightly reduces gamma (the option's curvature flattens as the probability distribution of outcomes widens) but increases theta (more time value to decay per day). The gamma-to-theta ratio typically worsens slightly.
- **Out-of-the-money converts:** Higher implied vol increases both gamma and delta, as the probability of the option finishing in the money rises. The convert becomes more "balanced" and potentially a better gamma candidate.
- **In-the-money converts:** Higher implied vol has modest effects since the option is already deep in the money.

34.3 Delta Shift and Rebalancing Implications

When implied volatility rises, the delta of the convertible increases (for OTM and ATM converts). This means the trader's existing short stock hedge is now insufficient — the position has become under-hedged. The trader must sell additional shares to re-establish delta neutrality. This forced selling after a

vol spike can be costly if the vol increase coincided with a stock decline (as is typical via the leverage effect), because the trader is selling into a declining, illiquid market.

KEY CONCEPT: Practical Implication of Rising Implied Vol

Rising implied vol is generally favorable for the gamma trader's mark-to-market but creates a rebalancing demand that can be expensive to execute. The trader profits on paper but must be careful that the hedge adjustment cost doesn't consume the vega gain. In practice, experienced traders may partially anticipate vol increases during market stress and pre-adjust hedges.

35. When Implied Volatility Falls

35.1 Direct Price Effect

A decline in implied volatility reduces the convertible's theoretical value. The option component becomes less valuable, and the convert's price falls. For a delta-hedged position, this is an unambiguous mark-to-market loss on the convertible leg. This is the vega bleed that many gamma traders underestimate.

The danger is particularly acute in two scenarios:

- **Post-catalyst vol crush:** If a stock had elevated implied vol due to an upcoming event (earnings, FDA decision, merger vote) and the event passes without a large move, implied vol collapses. The gamma trader who bought the convert at high implied vol suffers a double loss: the vol crush destroys mark-to-market value, and the lack of realized movement means little gamma was harvested to compensate.
- **Gradual implied vol erosion:** In a low-volatility environment, implied vol can drift downward over weeks or months as the market reprices expected future movement. The gamma trader bleeds slowly through vega without the large stock moves needed to generate gamma P&L.

35.2 The Implied-Realized Disconnect

This is perhaps the most subtle and important dynamic. It is entirely possible for realized volatility to exceed implied volatility (making the gamma harvest profitable in isolation) while implied volatility simultaneously declines (creating a vega loss that partially or fully offsets the gamma gains). This occurs when the market was initially pricing in even higher future vol that failed to materialize, even though current realized vol is adequate.

Formally, the total P&L of the delta-hedged position over a period can be decomposed as:

$$P\&L \approx \tfrac{1}{2}\Gamma S^2(\sigma^2_realized - \sigma^2_implied,entry) \times dt + Vega \times \Delta\sigma_implied + Carry$$

The first term is the gamma-theta component (positive when realized > implied at entry). The second term is the vega component (negative when implied vol falls). The third is the carry. A complete analysis must account for all three.

KEY CONCEPT: The Vol Surface Matters

The convertible's implied vol does not exist in isolation. It is influenced by the listed options implied volatility surface, the credit spread of the issuer, supply-demand technicals in the convertible market, and the general risk appetite environment. A gamma trader must understand what is driving implied vol changes to assess whether they are temporary (and thus tradeable) or structural (requiring position adjustment).

35.3 Effect on Gamma Profile

Falling implied vol has a particularly pernicious effect on the gamma profile of ATM converts: gamma actually increases as implied vol falls (the option's probability density concentrates near the strike), but theta also increases disproportionately. The breakeven move $\sqrt{(2\theta/\Gamma)}$ may remain roughly constant or

even decrease, but the position's mark-to-market has already suffered the vega loss. The trader is now harvesting gamma from a lower cost basis but has already taken a capital hit.

36. Implied Vol Term Structure and Roll-Down

Convertible bonds are long-dated instruments (typically 2–5 year options), and the term structure of implied volatility affects their behavior over time:

- **Contango (upward-sloping term structure):** When long-dated implied vol exceeds short-dated, the convertible benefits from "rolling down" the term structure as time passes. As the option's remaining life shortens, it moves to a lower point on the term structure, and if the curve is steep, this implicit repricing partially offsets time decay. This is favorable for the gamma trader.
- **Backwardation (downward-sloping term structure):** When short-dated vol exceeds long-dated (common during acute market stress), the roll-down works against the trader. As the option ages, it moves to a lower implied vol level, creating an additional headwind beyond normal theta decay.

Monitoring the equity options term structure on the same underlying provides insight into whether the convertible's implied vol is likely to drift up or down purely from the passage of time.

37. Implied Vol, Credit Spreads, and the Feedback Loop

For convertible bonds, implied volatility and credit spreads are not independent. They interact through a feedback loop that can amplify losses:

- Credit deterioration → stock price decline → implied vol spike (leverage effect) → delta drops → convert moves toward busted territory.
- In busted territory, the convert's price is driven primarily by the bond floor, which itself is falling due to widening credit spreads.
- The vega gain from the implied vol spike is overwhelmed by the credit-driven decline in the bond floor.

This is the "wrong-way" vega scenario: implied vol rises but the position loses money because the credit component dominates. It is one of the most dangerous dynamics in convertible bond arbitrage and has been extensively documented in the literature on convertible arbitrage during the 2008 financial crisis (Calamos, 2003; Woodson, 2002; Lhabitant, 2002; Mitchell & Pulvino, 2012).

KEY CONCEPT: Credit-Equity-Vol Nexus

In the words of practitioners, "you can't hedge credit with gamma." When the issuer's credit quality deteriorates severely, the convertible's behavior becomes dominated by credit risk regardless of what implied vol does. The gamma trader must maintain strict credit discipline and exit positions before the credit-equity-vol feedback loop activates.

38. Vega Hedging Strategies

Sophisticated convertible bond desks actively manage vega exposure rather than passively accepting it:

38.1 Listed Options Overlay

Selling listed equity options (typically puts or straddles) against the convertible position can offset some of the vega exposure. If the convertible is long vega at 3-year maturity, selling 6–12 month listed options creates a partial vega hedge. However, the maturity mismatch means the hedge is imperfect: short-dated implied vol can move differently from the long-dated implied vol embedded in the convert. This approach is most effective when the listed options are rich relative to the convertible (creating a "vol surface arbitrage" that captures the spread between listed and convertible implied vols).

38.2 Variance Swaps

Variance swaps provide a cleaner separation of realized and implied volatility exposure. By entering a variance swap (paying fixed, receiving realized variance), the trader can lock in a specific realized-implied spread independent of subsequent implied vol movements. However, variance swaps introduce their own risks (gap risk, convexity of the variance payoff) and are not available on all underlyings.

38.3 Portfolio-Level Vol Management

At the portfolio level, diversifying across many convertible bond positions with different underlying stocks provides natural vega diversification. While a portfolio-wide implied vol shock (e.g., a VIX spike or collapse) affects all positions, idiosyncratic implied vol moves across different names tend to partially cancel. Maintaining a target portfolio vega and adjusting position sizes to keep aggregate vega within risk limits is standard practice on institutional convertible desks.

39. Empirical Evidence and Notable Research

The interaction between implied volatility and convertible bond gamma trading has been studied extensively:

Author(s) / Source	Key Contribution
Calamos (2003), Convertible Arbitrage	Foundational text documenting the carry-gamma-vega decomposition and practical management of implied vol exposure in convertible portfolios
Woodson (2002), Global Convertible Investing	Detailed analysis of how changes in listed options IV propagate to convertible bond pricing and the resultant hedge adjustments
De Spiegeleer & Schoutens (2011)	Rigorous quantitative framework for isolating vega from gamma in convertible bond valuation, including the impact of stochastic vol models
Mitchell & Pulvino (2012), NBER	Empirical study of convertible arb returns during the 2008 crisis showing how credit-driven implied vol spikes destroyed gamma trade economics
Choi, Getmansky, Henderson & Tookes (2010)	Documented the feedback between convertible bond hedging activity and underlying stock volatility, showing that arb hedging itself can amplify vol
Agarwal, Fung, Loon & Naik (2011)	Decomposed convertible arbitrage returns into factor exposures including vega, demonstrating that implied vol changes explain a significant portion of return variance
Loncarski, ter Horst & Veld (2009)	Empirical analysis of the cheapness of convertible bonds relative to listed options across market regimes, finding systematic cheapness that varies with implied vol levels
Ammann, Kind & Wilde (2003)	Showed that convertible bond implied vols systematically understate listed option vols, creating a structural gamma advantage that fluctuates with market conditions

40. Practical Decision Framework for Implied Vol Scenarios

Implied Vol Scenario	Effect on Position	Recommended Action
IV rising, realized vol also high	Favorable: vega gain + gamma harvest	Maintain position; this is the ideal regime. Consider taking partial vega profit if IV reaches

Implied Vol Scenario	Effect on Position	Recommended Action
		extreme levels.
IV rising, realized vol low	Mixed: vega gain offsets lack of gamma P&L	Monitor closely. The IV rise may signal incoming realized vol. If IV rise is purely technical (e.g., pre-earnings), consider whether to hold through the event.
IV falling, realized vol high	Dangerous disconnect: gamma P&L offset by vega loss	Assess magnitude. If gamma harvest exceeds vega bleed, continue. If vega loss dominates, consider listed options overlay to cap further vega losses.
IV falling, realized vol low	Worst case: no gamma P&L, vega bleed, theta bleed	Reduce or exit. This is the regime where gamma trades die slowly. Redeploy capital to names with better cheapness signals.
IV spike from credit stress	Wrong-way risk: IV gain overwhelmed by credit loss	Exit immediately if credit thesis is broken. Do not be seduced by the apparent vega gain — the bond floor is collapsing beneath you.
IV crush post-catalyst	Sharp vega loss, possibly inadequate gamma harvest	Accept the loss if position-sized correctly. Avoid concentrated pre-catalyst positions unless the gamma opportunity during the event window compensates for the vol crush risk.

KEY CONCEPT: The Practitioner's Rule of Thumb

Experienced convertible bond gamma traders often say: "We trade realized vol but we're marked on implied vol." The most important implication is that position sizing must account for potential vega losses, not just the gamma-theta economics. A trade that is profitable on gamma-theta but destroys capital through vega can still be a bad trade if the vega loss exceeds the trader's risk tolerance or holding period.

Part VII — Non-Linear Transaction Costs and Thin Liquidity

18. Why the Standard Framework Breaks Down

The classical Whalley-Wilmott result assumes proportional transaction costs (cost = λ × n × S). In reality, transaction costs are almost never purely proportional. The actual cost consists of:

- Explicit spread cost: Bid-ask spread that widens with trade size and narrows with patience.
- Market impact cost: Price movement caused by trading — a concave function of trade size.
- Opportunity cost of delay: Risk of further adverse movement during slow execution.
- Information leakage: Market may infer your need to trade and move against you.

19. The Square-Root Impact Model

The most empirically robust market impact model (Almgren, Chriss, Bouchaud, Farmer, Lillo) is the square-root law. Temporary impact of trading n shares with ADV = V and volatility σ:

$$\Delta P \approx \sigma \cdot \eta \cdot \sqrt{n/V}$$

where η is a dimensionless constant typically 0.1–0.5. Total cost becomes:

$$C(n) = \eta\sigma \cdot n^{3/2} / \sqrt{V}$$

This n^(3/2) cost function is super-linear: large rebalancing trades are disproportionately expensive. This has profound implications for the optimal bandwidth.

20. Adjusted Optimal Bandwidth

Under square-root impact costs, the optimal band scales differently:

Cost Model	Bandwidth Scaling	Practical Implication
Proportional (Whalley-Wilmott)	$b^* \propto \lambda^{1/3}$	Classical cube-root; band insensitive to cost
Square-Root Impact	$b^* \propto (\text{cost param})^{2/7}$	Even less sensitive; narrower bands, more frequent smaller rebalances
General Power-Law $\lvert n\rvert^p$	$b^* \propto \kappa^{1/(p+1)}$	Higher convexity in costs → narrower bands

Because large trades are super-linearly expensive, the optimal policy involves more frequent but smaller rebalances than the proportional-cost policy would suggest.

21. Liquidity-Adjusted Bandwidth

When the underlying stock's liquidity is thin, additional complications arise:

- **Volume-dependent impact:** Impact ∝ √(n/V); when V is small, impact per share is large.

- **Intraday liquidity variation:** Liquidity is highest at open/close, thinnest midday. Bandwidth should be time-of-day-dependent.
- **Urgency-liquidity tradeoff:** Trading immediately minimizes time off-hedge but maximizes impact. Splitting orders reduces impact but extends hedge imperfection.

A practical heuristic: size each rebalancing trade to be no more than 5–10% of the stock's trailing 20-day ADV when executed over a single hour, and use execution algorithms (VWAP, TWAP, or IS algorithms) rather than market orders.

21.1 Liquidity Score Integration

Compute a real-time liquidity score incorporating current bid-ask spread, quoted depth, recent volume, order book imbalance, and estimated impact for a standardized trade. Adjust bandwidth as:

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b*_adjusted = b*_base × (L_average / L_current)^β
```

where β is typically 0.3–0.5. When liquidity is below average, the bandwidth widens; above average, it tightens.

Part VIII — The Liquidity Risk Premium

22. Core Insight: Sampling Bias in Rebalancing

The liquidity risk premium (LRP) stems from a fundamental sampling bias: the trader does not transact uniformly across market states. Rebalancing is triggered by large hedge-error accumulation, which happens fastest when volatility is high — precisely the periods when transaction costs are elevated. The trader therefore "samples" the cost distribution disproportionately in its expensive tail.

23. Joint Dynamics Setup

The stock price, volatility, and transaction costs evolve jointly:

- Stock: $dS/S = \mu dt + \sigma(t)dW_1$
- Volatility: $d\sigma = \kappa_\sigma(\bar{\sigma} - \sigma)dt + \eta dW_2$
- Transaction cost: $\lambda(t) = \lambda_0 + \beta(\sigma(t) - \bar{\sigma})$

where $\beta > 0$ captures the empirical regularity that spreads widen when volatility rises, and $\mathrm{corr}(dW_1, dW_2) = \rho_sv$ (the leverage effect, typically negative for equities).

24. Derivation of the LRP

The rebalancing frequency conditional on volatility state σ is:

$$f(\sigma) = 2\Gamma^2\sigma^2S^2 / b^2$$

This is proportional to σ^2: in high-volatility states, the trader rebalances quadratically more often. The effective average cost is not the unconditional average $\bar{\lambda}$ but the frequency-weighted average:

$$\lambda_eff = E[\lambda(\sigma) \cdot \sigma^2] / E[\sigma^2]$$

Applying the covariance decomposition $E[XY] = E[X]E[Y] + Cov(X,Y)$:

$$\lambda_eff = \bar{\lambda} + Cov(\lambda(\sigma), \sigma^2) / E[\sigma^2]$$

The LRP is therefore:

$$LRP = Cov(\lambda(\sigma), \sigma^2) / (\bar{\lambda} \cdot E[\sigma^2])$$

Since λ and σ^2 are positively correlated ($\beta > 0$), the LRP is strictly positive.

25. Explicit Evaluation (Linear Cost-Volatility Model)

Under $\lambda(\sigma) = \lambda_0 + \beta(\sigma - \bar{\sigma})$ with Ornstein-Uhlenbeck volatility (stationary Gaussian with mean $\bar{\sigma}$ and variance $v^2_\sigma = \eta^2/(2\kappa_\sigma)$):

$$LRP = 2(\beta/\lambda) \cdot \bar{\sigma} \cdot \varphi^2 / (1 + \varphi^2)$$

where $\varphi = v_\sigma/\bar{\sigma}$ is the coefficient of variation of volatility (relative vol-of-vol).

25.1 Numerical Example

Parameter	Value	Description
$\bar{\sigma}$	0.35	35% mean annualized vol

Parameter	Value	Description
φ	0.30	30% relative vol-of-vol
$\beta/\bar{\lambda}$	1.5	10% vol increase → 15% cost increase
LRP	≈8.7%	Effective cost ~9% above unconditional average

For illiquid names where $\beta/\bar{\lambda}$ is 3–5 or during regime shifts with elevated φ, the premium can reach 20–40%.

26. Complete Adjusted Bandwidth Formula

Incorporating both the LRP (higher effective costs) and effective volatility correction (higher effective σ):

$$b^*_{adj} = (3\lambda(1+LRP) / (2\Gamma\sigma^2(1+\varphi^2)))^{(1/3)}$$

The ratio to the naive bandwidth is:

$$b^*_{adj} / b^*_{naive} = ((1+LRP) / (1+\varphi^2))^{(1/3)}$$

The numerator widens the band (higher effective costs), the denominator narrows it (higher effective volatility). When cost sensitivity to volatility is high (illiquid names), the LRP dominates and the band widens. When vol-of-vol is high but cost sensitivity moderate, the effective-volatility correction dominates and the band narrows.

Part IX — Leverage Effect and Asymmetric Rebalancing Boundaries

27. The Symmetry-Breaking Mechanism

The leverage effect (negative correlation between equity returns and volatility innovations, $\rho_{sv} < 0$) enters the hedger's HJB equation through the cross-diffusion term $\rho_{sv} \cdot c(\sigma) \cdot \eta \cdot V_{e\sigma}$. This is the only term that is odd in the hedge error e when all other terms are even, making it the sole source of bilateral symmetry breaking.

Decomposing the value function $V(e,\sigma) = V^{(0)}(e,\sigma) + \rho_{sv} \cdot V^{(1)}(e,\sigma) + O(\rho^2_{sv})$, the zeroth-order term is even in e (standard Whalley-Wilmott) while the first-order correction $V^{(1)}$ is forced to be odd. An odd correction to an even value function shifts the minimum and distorts the boundary asymmetrically.

28. Band Shift Rather Than Widening

The no-trade region $[-b_-, b_+]$ is shifted at first order: the total width is preserved, but the center moves. With $\rho_{sv} < 0$:

- **Upper boundary (long delta) widens:** $b_+ = b_0 + |\rho_{sv}| \cdot |\delta| > b_0$
- **Lower boundary (short delta) tightens:** $b_- = b_0 - |\rho_{sv}| \cdot |\delta| < b_0$

The fractional asymmetry is expressed through the dimensionless parameter:

$$\xi = (b_+ - b_-) / (b_+ + b_-) = |\rho_{sv}| \cdot \eta\beta / (\alpha\sigma) \cdot S \cdot (2\Gamma\sigma^2/3\lambda)^{(2/3)}$$

29. Two Distinct Channels of Asymmetry

29.1 Cost-Correlation Channel (Dominant, $O(\rho_{sv})$)

Hitting the short-delta boundary requires $dW_1 < 0$ (stock decline). With $\rho_{sv} < 0$, this coincides with a volatility spike, so $\lambda(\sigma)$ is elevated at the exact moment of the transaction. Conversely, hitting the upper boundary coincides with vol decline and cheaper execution. This channel is proportional to β and vanishes when transaction costs are volatility-independent.

29.2 Running-Cost Channel (Second-Order, $O(\eta^2)$)

Even with constant costs ($\beta = 0$), the leverage effect creates asymmetry through the state-dependent running cost $\frac{1}{2}\alpha\sigma^2e^2$. When short delta and the stock falls further, volatility rises and amplifies the running cost. This positive feedback loop enters the boundary correction only at $O(\eta^2)$.

30. The Self-Reinforcing Adverse Spiral

Starting from $e < 0$ (short delta) with $dS < 0$:

- Hedge error becomes more negative ($de = \Gamma\sigma S\, dW_1 < 0$)
- Volatility jumps ($d\sigma > 0$ via leverage effect)
- Transaction costs spike ($d\lambda = \beta\, d\sigma > 0$)

Each change worsens the position: larger error, faster accumulation rate, higher cost to rectify. This is a self-reinforcing adverse spiral.

Compare to the long-delta case: e > 0, dS > 0 causes error growth, but volatility drops and costs decline. The higher error is partially offset by a more benign environment. This is self-correcting.

KEY CONCEPT: The Optimal Response

The tighter short-delta boundary is the HJB equation's way of expressing: the expected cost of tolerating an additional unit of short-delta error, accounting for correlated deterioration in volatility and costs, exceeds the expected cost of tolerating long-delta error.

31. Quantitative Magnitudes

Market Regime	ξ Range	Example (symmetric band ±100 shares)
Liquid convertibles, calm markets	0.03–0.08	−95 to +105
Illiquid small-caps, normal vol	0.08–0.15	−90 to +110
Illiquid, stressed markets	0.15–0.20	−85 to +115

Key sensitivities:

- Asymmetry scales linearly with |ρ_sv| at leading order. The leverage effect is regime-dependent: |ρ_sv| ranges from ~0.4 in calm markets to 0.85+ during selloffs.
- Entire asymmetry reverses for negative gamma (short converts or written warrants).
- A static symmetric band calibrated in calm periods will systematically under-rebalance short-delta positions during drawdowns.

Part X — The Gamma Trade Lifecycle

32. End-to-End Workflow

Step 1: Screening

Screen the universe for balanced converts with delta 40–60%, maturity 2–5 years, good credit, liquid and volatile underlying stocks, low borrow costs, and clean structures.

Step 2: Cheapness Assessment

Use model pricing, implied vol comparison against listed options and realized vol, and practical market indicators to filter for converts where embedded gamma is offered at a discount to fair value.

Step 3: Trade Construction

Buy the convert, short delta-equivalent shares, and calculate the initial carry profile (coupon + short rebate − funding − borrow). Establish rebalancing parameters — typically a delta band of 3–5 points for liquid names, with end-of-day time-based rebalancing.

Step 4: Ongoing Management

Monitor implied-realized spread, gamma-to-theta ratio, credit health, borrow availability, and moneyness migration. Execute rebalancing systematically, increasing frequency in high-vol regimes and decreasing in quiet markets. Dynamically adjust position size as cheapness evolves.

Step 5: Exit Criteria

Exit when the convert migrates out of the gamma-rich zone (busted or deeply ITM), when cheapness disappears or reverses, when credit fundamentals deteriorate, when borrow costs become prohibitive, or when the convert approaches maturity or a call date that truncates remaining optionality.

> **KEY CONCEPT: Discipline**
> The most disciplined gamma traders treat exit signals with the same rigor as entry signals, recognizing that gamma trading in convertibles is a strategy of continuous optimization rather than a single bet.

Part XI — Quick Reference: Key Formulas

Formula	Expression	Meaning
Gamma P&L (single rebalance)	$0.5 \times \Gamma \times (\Delta S)^2$	Convexity gain from rebalancing; always positive
Net Carry	Coupon + Short Rebate − Funding − Borrow	Income earned while holding the position
Breakeven Daily Move	$\sqrt{(2\theta/\Gamma)}$	Stock move needed per day to offset theta
Whalley-Wilmott Band	$(3\lambda / 2\Gamma\sigma^2)^{(1/3)}$	Optimal no-trade bandwidth (proportional costs)
Square-Root Impact	$\Delta P \approx \sigma\eta\sqrt{(n/V)}$	Market impact of trading n shares
Liquidity Risk Premium	$\mathrm{Cov}(\lambda,\sigma^2) / (\bar{\lambda} \cdot E[\sigma^2])$	Cost markup from vol-cost correlation
LRP (explicit)	$2(\beta/\bar{\lambda})\bar{\sigma}\varphi^2 / (1+\varphi^2)$	For linear cost-volatility model with OU vol
Adjusted Bandwidth	$(3\bar{\lambda}(1+\mathrm{LRP}) / 2\Gamma\bar{\sigma}^2(1+\varphi^2))^{(1/3)}$	Incorporates LRP and effective vol
Fractional Asymmetry	$\lvert\rho_{sv}\rvert\eta\beta S / (\alpha\sigma) \cdot (2\Gamma\sigma^2/3\lambda)^{(2/3)}$	Leverage-induced boundary asymmetry
Liquidity-Adjusted Band	$b^*_{base} \times (L_{avg}/L_{current})^{\beta}$	Dynamic bandwidth based on liquidity score

Part XII — Key Variable Definitions

Symbol	Definition
Γ (Gamma)	Second derivative of option price w.r.t. stock price; measures delta sensitivity
Δ (Delta)	Sensitivity of convertible price to stock price; hedge ratio
Vega	Sensitivity of convertible price to changes in implied volatility
θ (Theta)	Time decay of the embedded option per day
σ	Instantaneous or realized volatility of the underlying stock
$\bar{\sigma}$	Long-run mean volatility
η	Vol-of-vol parameter (volatility of the volatility process)
φ	Relative vol-of-vol: $v_\sigma / \bar{\sigma}$ (coefficient of variation)
λ	Proportional transaction cost parameter
β	Sensitivity of transaction costs to volatility: $d\lambda/d\sigma$
ρ_{sv}	Correlation between stock returns and vol innovations (leverage effect)
b^*, b_0	Optimal half-width of the no-trade band
b_+, b_-	Upper (long delta) and lower (short delta) boundaries

Symbol	Definition
ξ	Dimensionless fractional asymmetry measure
$e(t)$	Hedge error: theoretical delta minus actual hedge ratio
LRP	Liquidity risk premium: effective cost markup from vol-cost covariance
$V(e,\sigma)$	Value function in the HJB equation (minimum expected total cost)
L_σ	Infinitesimal generator of the volatility diffusion process
α	Risk-aversion / position-size parameter in running cost
S	Current stock price
V (volume)	Average daily trading volume of the stock